

RS004 · Platformer Game

Replit / Python — Side-scroller with gravity and collisions.

For parents: these are the key English coding words your child meets in this course, week by week. You don't need to code — just read a word together and ask your child to explain it back in their own words. The tiny example shows what it looks like in real code.

W1

WORD	WHAT IT MEANS	TINY EXAMPLE
<code>pygame.init()</code>	wake up the pygame engine	<pre>pygame.init()</pre>
<code>display.set_mode</code>	open a game window of a given size	<pre>pygame.display.set_mode((640, 480))</pre>
<code>clock.tick(60)</code>	limit the game to 60 frames per second	<pre>clock.tick(60)</pre>

W2

WORD	WHAT IT MEANS	TINY EXAMPLE
<code>sprite</code>	a movable character or object	<pre>player = pygame.Rect(50, 50, 32, 32)</pre>
<code>rect</code>	a rectangle that tracks position and size	<pre>player.x += 5</pre>
<code>delta</code>	the amount to move per frame	<pre>dx = 5; player.x += dx</pre>

W3

WORD	WHAT IT MEANS	TINY EXAMPLE
<code>gravity</code>	a constant downward pull each frame	<pre>vel_y += gravity</pre>
<code>velocity_y</code>	the current vertical speed	<pre>player.y += vel_y</pre>
<code>jump impulse</code>	a sudden upward velocity from a jump	<pre>vel_y = -12</pre>
<code>ground</code>	the surface that stops downward motion	<pre>if player.y > 400: player.y = 400</pre>

W4

WORD	WHAT IT MEANS	TINY EXAMPLE
<code>collision</code>	two rectangles overlap	<pre>player.colliderect(coin)</pre>
<code>colliderect</code>	pygame method that detects overlap	<pre>if a.colliderect(b):</pre>
<code>platform</code>	a surface the player can stand on	<pre>platforms = [pygame.Rect(...)]</pre>

W5

WORD	WHAT IT MEANS	TINY EXAMPLE
level	the layout of one stage	<pre>level = ["..#..", "#####"]</pre>
tile	one piece of a level grid	<pre>if cell == "#": draw_block()</pre>
coordinates	the (x, y) location on screen	<pre>pos = (x, y)</pre>

W6

WORD	WHAT IT MEANS	TINY EXAMPLE
collectible	an item the player picks up	<pre>coins = [pygame.Rect(...)]</pre>
score	the number that tracks progress	<pre>score += 10</pre>
remove	delete an item from a list once collected	<pre>coins.remove(coin)</pre>

W7

WORD	WHAT IT MEANS	TINY EXAMPLE
enemy	a sprite that hurts the player on contact	<pre>if player.colliderect(enemy): die()</pre>
game state	playing / dead / won — the current mode	<pre>state = "playing"</pre>
respawn	put the player back at the start	<pre>player.x, player.y = 50, 50</pre>

W8

WORD	WHAT IT MEANS	TINY EXAMPLE
level transition	moving from one level to the next	<pre>level_index += 1</pre>
level list	list of level layouts to load in order	<pre>levels = [level1, level2]</pre>
polish	final touch-ups before shipping	<pre># add sound, fix bugs</pre>